

■ Machine Learning Formula Handbook (Running PDF)

■■ Date: 08/12/2025 – Linear Regression & Gradient Descent

Regression equation, MSE, gradient descent updates.

■■ Date: 09/12/2025 – Tree, SVM, Logistic, Scaling

Entropy, Gini, Sigmoid, MinMax scaling.

■■ Date: 10/12/2025 – Naive Bayes

Bayes theorem, Gaussian distribution.

■■ Date: 13/12/2025 – Kernel Functions

Linear, Polynomial, RBF, Sigmoid kernels.

■■ Date: 13/12/2025 – Regularization & Lasso

L1 and L2 regularization objective functions.

■■ Date: 13/12/2025 – KNN

Distance metrics: Euclidean, Manhattan, Minkowski.

■■ Date: 13/12/2025 – Classification Metrics

Precision, Recall, F1-score formulas.

■■ Date: 13/12/2025 – PCA

Covariance matrix, eigenvalues, eigenvectors.

■■ Date: 13/12/2025 – Bias & Variance

Error decomposition concept and trade-off.

■■ Date: 13/12/2025 – Bootstrap Aggregation (Bagging)

Bootstrap sampling and aggregation logic.